

KKchemstruct User Manual

KKTeX

2 August 2026 v1.0.0

目次

1	Overview	2
1.1	Requirements	2
1.2	Loading	2
1.3	When two compilations are needed	2
2	Benzene rings	2
2.1	<code>\KKbenzene</code>	2
2.2	Options	3
2.3	Rings set inline	3
2.4	Fused rings (<code>\KKchemring</code>)	3
3	Condensed formulas	4
3.1	<code>\KKchemchain</code>	4
3.2	Common fragments	4
4	Reaction schemes	4
4.1	<code>\KKscheme</code>	4
4.2	<code>\KKbranchscheme</code>	5
4.3	Compound names (<code>\KKname</code>)	5
5	Annotations	5
5.1	Marking a leaving group (<code>\KKchemmark</code>)	6
5.2	Hydrogen bonds (<code>\KKhbond</code>)	6
5.3	Crossing out (<code>\KKchemcross</code>)	6
6	Chemical formula macros	6
7	Changing the dimensions	6
7.1	<code>\KKchemsetup</code>	7
7.2	List of keys	7

8	Predefined compounds ·····	7
9	Pitfalls ·····	8
10	License ·····	8

1 Overview

KKchemstruct is a set of macros for typesetting the structural formulas of high-school chemistry (organic and inorganic). The drawing itself is done by `chemfig`; this package configures its dimensions, line widths and orientations to match the look of printed textbook figures, and adds a short notation for substituted benzenes, condensed formulas, reaction schemes and annotations.

Every dimension is defined in `em`, so the figures follow the body font size. There is no need to rewrite the numbers for each paper size.

1.1 Requirements

- Engine: LuaLaTeX, or pLaTeX + dvipdfmx
- Packages: `chemfig`, `etoolbox`, `xkeyval` (all part of TeX Live)

1.2 Loading

Put `KKchemstruct.sty` on the TeX search path or next to your document and write

```
\usepackage{KKchemstruct}
```

There are no package options.

1.3 When two compilations are needed

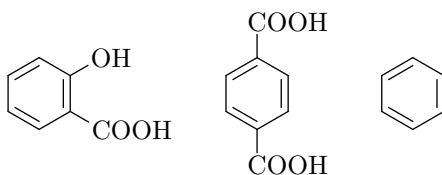
`\KKchemmark` and `\KKhbond` refer to TikZ nodes, so they need two compilations to settle. On the first pass the frame or the dotted line appears in the wrong place; the second pass puts it right.

2 Benzene rings

2.1 \KKbenzene

Substituted benzenes are written with `\KKbenzene`. Positions are numbered 1 to 6 clockwise, starting from the top vertex.

```
\KKbenzene[2=OH,3=COOH]
```



Substituents are set in math mode, so subscripts such as `COOCH3` work as written. Positions you leave out produce no branch at all.

2.2 Options

`1-6` the substituent at each position.

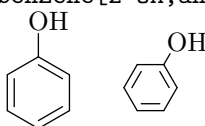
angle rotation of the whole ring, in degrees. The default is no rotation (vertices at top and bottom).

sep the side length (distance between atom centres) of this ring only.

kekule=b flips the inner double-bond strokes to the upper-right, bottom and upper-left sides. Use it when a substituent collides with a stroke.

`\KKbenzene[1=OH,kekule=b]`

`\KKbenzene[2=OH,angle=30,sep=1.0em]`



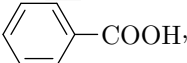
2.3 Rings set inline

To put a ring in running text, use the flat-top form, whose top and bottom sides are horizontal. It bonds at the left and right vertices, so it sits naturally in the middle of a chain.

`\KKphenylring` the bare ring.

`\KKphenyl{...}` a ring with a substituent to its right.

`\KKphenylene{...}` adds a bond stub on the left as well, so the ring can be inserted into a chain.

A ring such as  in running text does not disturb the leading. A substituent can be attached on the right, as in  and a ring can be dropped into a chain: $\text{CH}_3(\text{CH}_2)_{11}$ —— SO_3H .

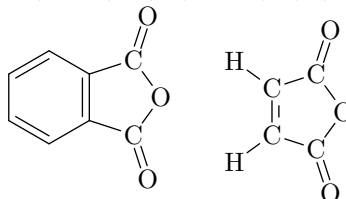
The amount by which such rings are raised above the baseline is set with the `inline raise` key:

`\KKchemsetup{inline raise=0.35ex}`

2.4 Fused rings (`\KKchemring`)

For rings other than benzene, and for fused rings, pass raw `chemfig` notation to `\KKchemring`. Only the ring dimensions (constant distance between atom centres) are applied. The optional first argument takes `chemfig` keys.

`\KKchemring[atom sep=1.75em]{*6(=*5(-C(=O)-O-C(=O)-)=--=)}`



Rings that carry C or O labels on their vertices need a longer side than skeleton-only rings, or the

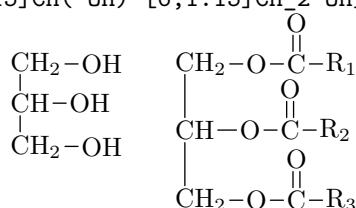
labels crowd each other. The default for those is `label ring sep (1.75em)`.

3 Condensed formulas

3.1 `\KKchemchain`

Formulas in which labels run horizontally, such as `CH_2-O-C(=[2]O)-R_1`, are written with `\KKchemchain`. Here the *visible* bond length is kept constant, so the line is not squeezed after a wide label such as `CH_2`.

`\KKchemchain{CH_2(-OH)-[6,1.15]CH(-OH)-[6,1.15]CH_2-OH}`



Angles use `chemfig` numbering in steps of 45 degrees (0 = right, 2 = up, 4 = left, 6 = down). The number after the angle is a length coefficient.

3.2 Common fragments

Ready-made pieces for use inside a chain. They are `chemfig` code fragments, so they belong in the argument of `\KKchemchain`.

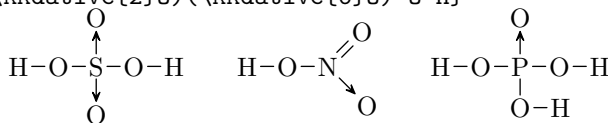
`\KKcarbonyl` an upward $C=O$.

`\KKcarbonyldown` a downward $C=O$.

`\KKdative{<angle>}` a dative bond arrow ($S \rightarrow O$, $P \rightarrow O$, $N \rightarrow O$).

Vertical bonds are drawn longer than horizontal ones: without allowing for the height of the labels and for the arrowhead, they look cramped.

`\KKchemchain{H-O-S(\KKdative{2}O)(\KKdative{6}O)-O-H}`



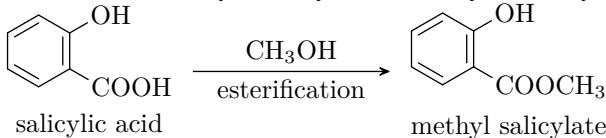
4 Reaction schemes

4.1 `\KKscheme`

A scheme is written by putting `chemfig` `\arrow` commands into the argument of `\KKscheme`. The arrow syntax is `chemfig`'s own.

`\KKscheme{%`

```
\KKname{\KKsalicylicacid}{salicylic acid}
\arrow{->[\ck{CH_3OH}][esterification]}
\KKname{\KKmethylsalicylate}{methyl salicylate}
```



The optional first argument takes `\schemestart` keys (`arrow angle`, `arrow coeff`, `arrow style`).

The default distance between compounds, `compound sep`, is 5.5em, which suits short labels such as chemical formulas. A long word set over the arrow – `esterification` above – is wider than that, and would run into the compound on its left; the example is therefore wrapped in

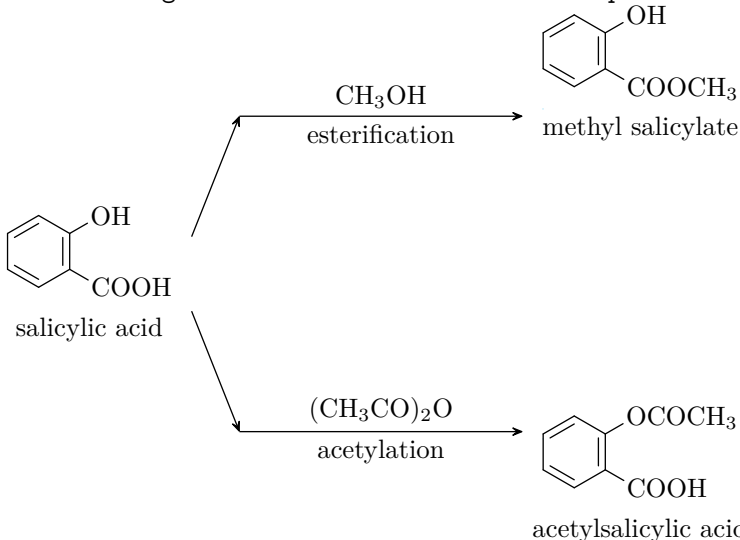
```
\begingroup\KKchemsetup{compound sep=9em} ... \endgroup
```

4.2 \KKbranchscheme

A figure in which one substrate branches into two products is written with `\KKbranchscheme`, which takes seven arguments. Each arrow rises at an angle and then runs horizontally; the labels sit above and below the horizontal part.

```
\KKbranchscheme
```

```
{<substrate>}
{<upper reagent>}{<upper reaction>}{<upper product>}
{<lower reagent>}{<lower reaction>}{<lower product>}
```



The angle and the length of the branches are set with `branch angle` (default 20 degrees) and `branch coeff` (default 2.7).

4.3 Compound names (\KKname)

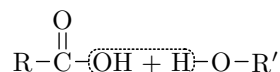
```
\KKname{<structure>}{<name>} centres a name under a structure. The font is set with name font.
```

5 Annotations

5.1 Marking a leaving group (`\KKchemmark`)

The -OH and -H lost in a condensation can be enclosed in a dotted rounded frame. Tag the atoms with chemfig's `@{name}` and give the two tags as opposite corners.

```
\KKchemchain{R-C\KKcarbonyl-@{s}OH}\ +\ \KKchemchain{@{e}H-O-R'}%
\KKchemmark{s}{e}
```

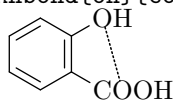


The optional first argument takes TikZ keys, for example to change the colour.

5.2 Hydrogen bonds (`\KKhbond`)

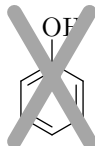
Draws the dotted line of an intramolecular hydrogen bond between two tags.

```
\KKbenzene[2=@{oh}OH,3=C@{co}OOH]\KKhbond{oh}{co}
```



5.3 Crossing out (`\KKchemcross`)

Superimposes a heavy cross to show that a reaction does not take place. The optional first argument sets the colour (default `black!35`).



6 Chemical formula macros

`chemfig` sets atom labels in math mode, so subscripted formulas such as CH_3 work as written inside a structure. Arrow labels, however, are not in math mode – that is how `chemfig` works. Wrap those in one of the following.

`\KKchemformula{...}` sets a formula with `\mathrm`.

`\ck{...}` a short alias for the above. It exists only to save keystrokes, so use `\KKchemformula` in documents where a two-letter name would clash.

`\ckm` / `\ckp` the superscript $-$ and $+$ of an ion.

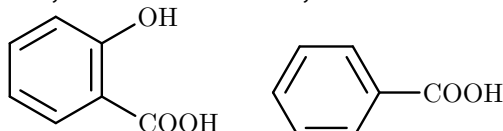
Use them as in CH_3COOH , RCOO^- and Na^+ . Labels in prose need no wrapper.

7 Changing the dimensions

7.1 \KKchemsetup

All dimensions are changed through `\KKchemsetup`.

```
\KKchemsetup{ring sep=2.0em,bond width=0.09em,double bond sep=0.4em}
```



Call it inside a group and the change stays in that group. The example above is wrapped in

`\begingroup ... \endgroup`, so here the original dimensions are back in force:

7.2 List of keys

ring sep (1.35em) side of a skeleton-only ring (distance between atom centres).

label ring sep (1.75em) side of a ring whose vertices carry C or O labels.

chain sep (0.75em) bond length in condensed formulas; the visible line is shorter than this by the gap at each end.

vertical coeff (1.15) how much vertical bonds are lengthened.

dative coeff (1.5) how much dative-bond arrows are lengthened.

mid bond coeff (1.28) coefficient of the middle horizontal bond in a triglyceride.

bond offset (0.13em) gap between a label and a bond line.

double bond sep (0.26em) distance between the two lines of a double bond.

bond width (0.05em) thickness of bond lines.

arrow label sep (0.30em) gap between a reaction arrow and its label.

arrow offset (0.8em) gap between a compound and the end of an arrow.

compound sep (5.5em) distance between compounds in a scheme.

kink (1.8em) the rise of a branching arrow.

branch angle (20) / branch coeff (2.7) angle and length of branching arrows.

inline raise (0.35ex) how far an inline ring is raised.

name font (\normalsize) font of the name set by `\KKname`.

8 Predefined compounds

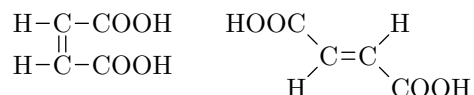
Frequently used structures can be called directly. Each is just a combination of the macros described above, so you can add your own by following the pattern at the end of the `.sty` file.

Aromatics `\KKsalicylicacid`, `\KKmethysalicylate`, `\KKaspirin`, `\KKbenzoicacid`, `\KKphenol`,
`\KKphthalicacid`, `\KKisophthalicacid`, `\KKterephthalicacid`

Anhydrides and geometric isomers `\KKphthalicanhydride`, `\KKmaleicanhydride`,
`\KKmaleicacid`, `\KKfumaricacid`

Oxoacids `\KKnitricacid`, `\KKsulfuricacid`, `\KKphosphoricacid`

Fats `\KKglycerol`, `\KKtriglyceride{R_1}{R_2}{R_3}`, `\KKestergeneric`



9 Pitfalls

Parentheses make branches In `chemfig`, (and) are the branch notation. Passing something like `CH_3(CH_2)_{11}` straight to `\KKchemchain` makes it a branch. Write formulas that are not drawn as structures with `\ck{...}` instead.

Arrow labels are not in math mode As above, wrap them in `\ck{...}`.

Annotations need two passes `\KKchemmark` and `\KKhbond` are not positioned on the first compilation.

Rings and chains fix different lengths Rings keep the distance between atom centres constant, which is what makes a regular hexagon; chains keep the visible line length constant, so wide labels do not squeeze the line. Use `\KKbenzene` / `\KKchemring` for rings and `\KKchemchain` for chains. Mixing them is safe: each macro switches the dimensions itself.

10 License

MIT License. See `LICENSE.md` in the repository for the full text.